Introduction to System Development

Katerina Mangaroska week 36

Learning Objectives

01 Understand the purpose of systems analysis and design in the development process.

02 The characteristics of iterative systems development.

03 Understand the six core processes of systems development life cycle.

04 The key documents used in the planning of a project.

ARCHITECT



PROGRAMMER

BUILDER



SYSTEM ANALYST

PRODUCT OWNER

BUYER



What is the basic purpose of a course in systems analysis and design?

(1) **understand and document** the business need, i.e., requirements,

(2) define and design a **solution**,

(3) **work in a team** to build the solution and,

(4) implement a solution that is **use-inspired.**

SYSTEM ANALYSIS AND DESIGN

03

providing the tools and techniques to understand the need (business need), capture the vision, define a solution, build the solution and lead others in building the solution, confirm that the solution meets the need, and launch the solution.

→ “soft” vs “hard” skills

SYSTEM ANALYSIS AND DESIGN

03

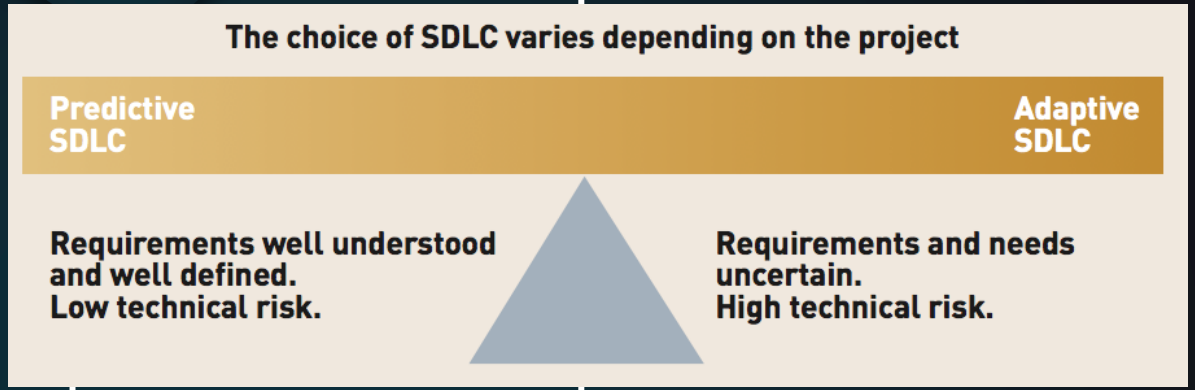
- activities that enable a person to understand and specify what the new system should accomplish.
  → what the system must do
- activities that enable a person to describe in detail the system that solves the needs.
  → how the system will work

SYSTEM ANALYST RESPONSIBILITIES

Research and understand the problem	Verify that the benefits of solving the problem outweigh the costs	Define the requirements for solving the problem
Develop a set of possible solutions (alternatives)	Decide which solution is best, and make a recommendation	Implement and monitor the solution

02

APPROACHES TO SOFTWARE DEVELOPMENT



PREDICTIVE

The project can be planned and organized in advance and developed according to the plan.

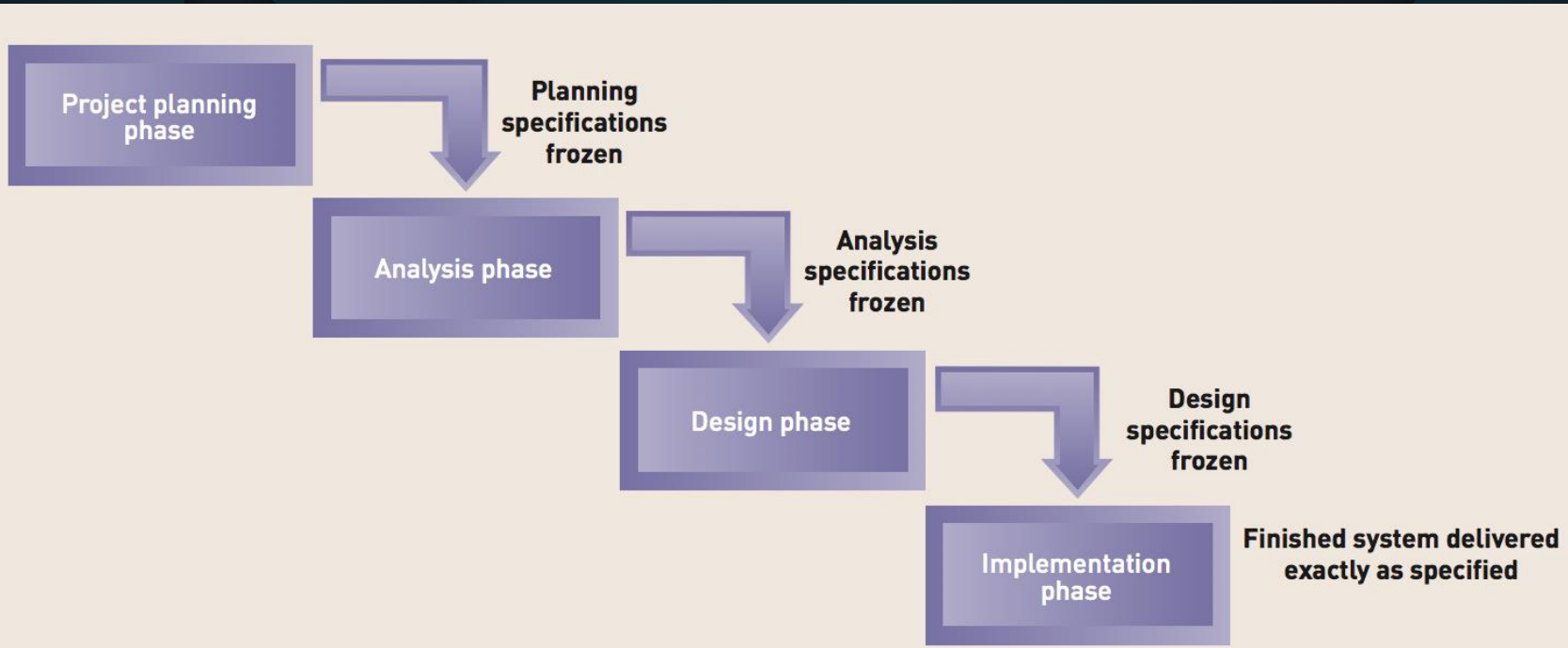
e.g., old mainframe inventory system convert to a newer networked client/server system.

The exact requirements of a system or the users' needs are not well understood. In this situation, the project cannot be planned completely in advance.

ADAPTIVE

WATERFALL METHODOLOGY

an SDLC approach that assumes the various phases of a project can be completed sequentially— one phase leads (falls) into the next phase

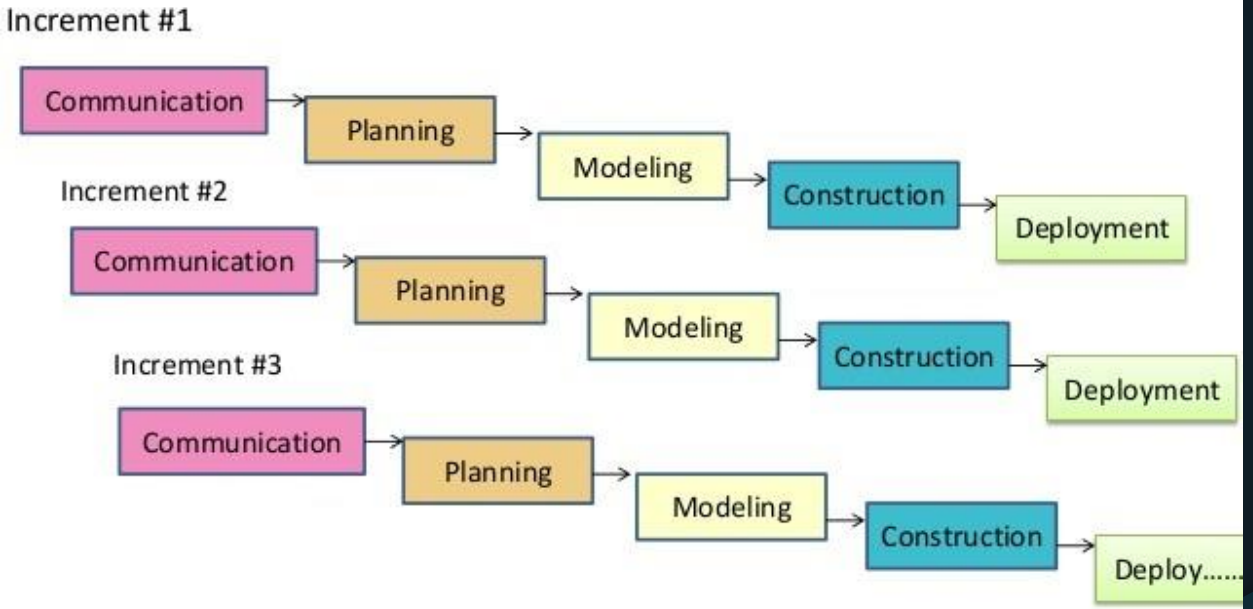

Project planning phase
Planning specifications frozen
Analysis phase
Analysis specifications frozen
Design phase
Design specifications frozen
Implementation phase
Finished system delivered exactly as specified

INCREMENTAL DEVELOPMENT

a development approach that completes parts of a system in several iterations and then puts them into operation for users

- What is the major difference with agile?

Incremental Model (Diagram)

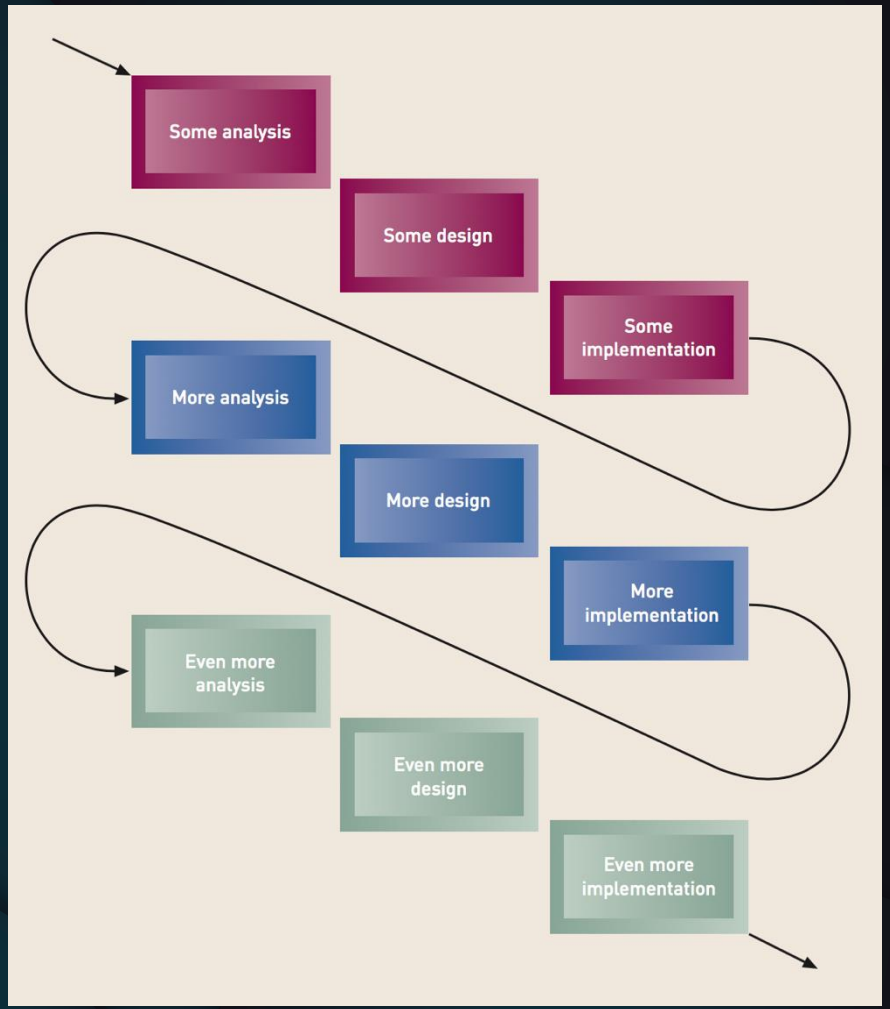


AGILE METHODOLOGY

information systems development process that emphasizes flexibility to anticipate new requirements during development

- Poor resource planning
- Limited documentation
- Fragmented output
- Difficult measurement
- No finite end

→ When is it enough?



The problem and solution spaces

How would you describe the way the waterfall process spans the two? What about iterative and incremental development processes?

03

SYSTEM DEVELOPMENT LIFE CYCLE (SDLC)

Includes all the activities required to build, launch, and maintain an information system.

Six core processes:

1. Identify the problem or need and obtain approval to proceed.
2. Plan and monitor the project—what to do, how to do it, and who does it.
3. Discover and understand the details of the problem or the need.
4. Design the system components that solve the problem or satisfy the need.
5. Build, test, and integrate system components.
6. Complete system tests and then deploy the solution.



Example through a retail company

To keep its product line current and popular, retail purchasing agents attend apparel and fabric trade shows around the world.

1980-1990

MAIL ORDER -- catalogs with mail-in and telephone orders – revenue **$10 million**

1994

brick- and-mortar – local stores – revenue **$67 million**

2000

online sales – web-based -- revenue **$200 million**

Trade shows

To keep its product line current and popular, retail purchasing agents attend apparel and fabric trade shows around the world. To place an order, they would exchange contact information with the seller at the trade show and upon returning to the home office would then follow up via e-mails and phone calls to formulate a contract and make a purchase order.

SYSTEM REQUIREMENTS

1. Collecting and storing information about the manufacturers/suppliers.
2. Collecting and storing information about sales representatives and other key personnel.
3. Collecting and storing information about products.
4. Collecting and storing images about products (or upload images).
5. Functioning as a stand-alone application without connection.
6. Connecting via Internet and transmitting data.

BUSINESS BENEFITS

1. Increase timely communication between trade show attendees and their home offices.

   → Improving the quality and speed of purchase order decisions.

2. Maintain correct and current information about suppliers and their representatives.

   → Facilitate rapid communication with suppliers.

3. Maintain correct and current information about new products.

   → Facilitate developments of catalogues and web pages.

4. Expedite the placing of purchase orders for new merchandise.

   → Keep up with trends and speed up product availability.

Fundamental development activities

Analysis involves understanding the problem which the software is intended to solve, i.e., the requirements in context, with **validation** as the means to check that understanding.

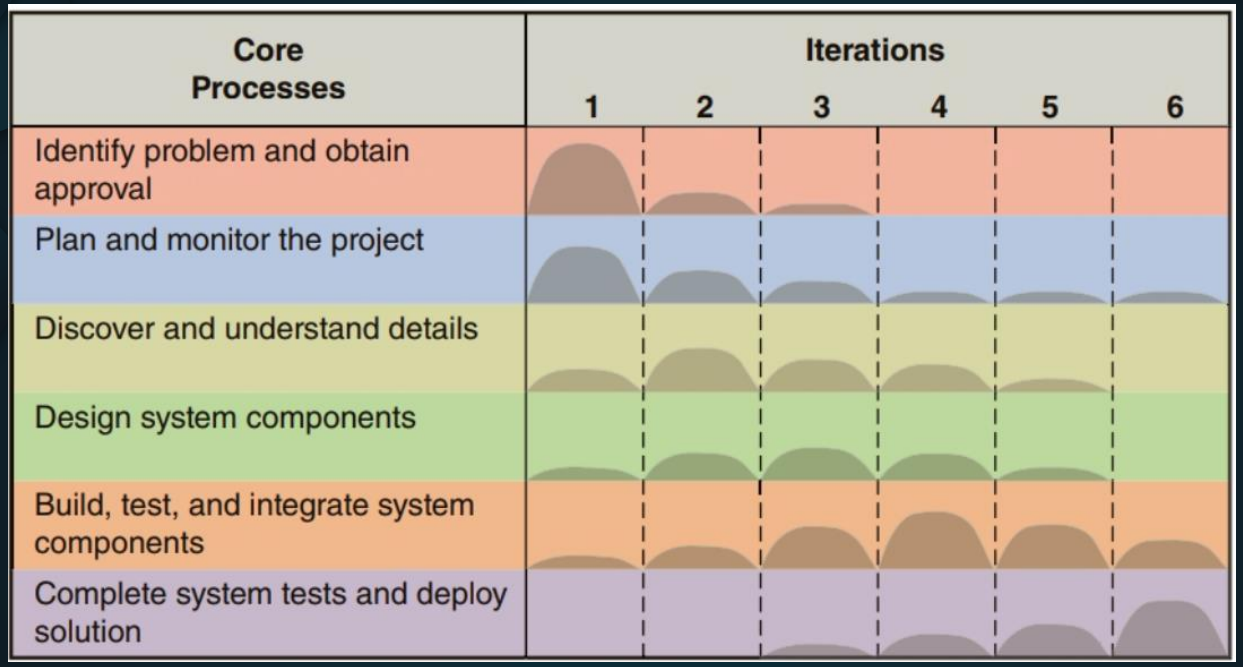
Design involves describing, conceptually, a software solution that meets the requirements of the problem.

Implementation involves realising such a solution in software.

Testing involves making sure that the solution has certain inherent qualities, with **verification** to check its adequacy with respect to the specified requirements and **validation** to check that the solution does address the problem.

Deployment involves making the developed solution available in its context of operation and use.

Iterative development



…is an approach to system development in which the system is “grown” in an almost organic fashion.

1. Identify the problem or need and obtain approval to proceed.

Review the system vision and scope documents of the project.

Determine the major components (functional areas) that are needed.

→ Supplier information system & Product information system (subsystems of CRM) in the example in slide 23.

2. Plan and monitor the project—what to do, how to do it, and who does it.

Define the iterations: 1. tasks per iteration, 2. sequence the tasks, 3. identify resources

Define team members roles and responsibilities.

Support tools:

Work Breakdown Structure (WBS) – list of tasks to do

Work sequence diagram (WSD) – visualize the tasks, dependences, and resources assigned

Work Breakdown Structure

I. Discover and understand the details of all aspects of the problem.

1. Meet with the Purchasing Department manager. ~ 3 hours
2. Meet with several purchasing agents. ~ 4 hours
3. Identify and define use cases. ~ 3 hours
4. Identify and define information requirements. ~ 2 hours
5. Develop workflows and descriptions for the use cases. ~ 6 hours

II. Design the components of the solution to the problem.

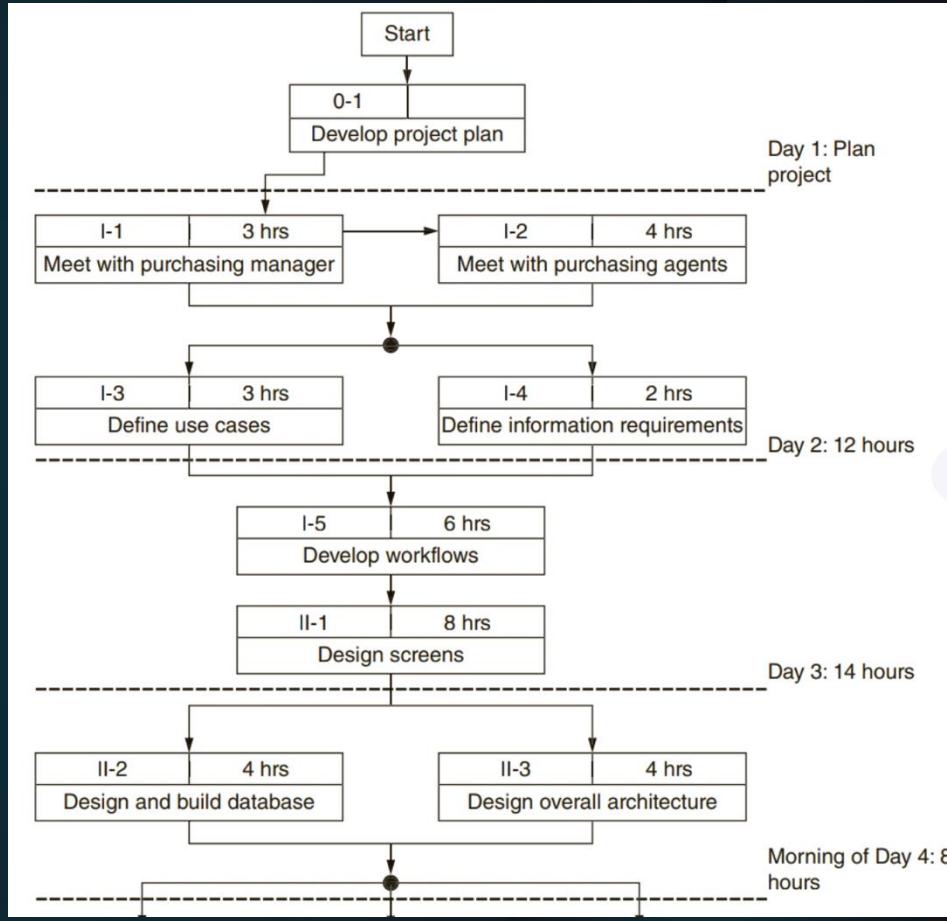
1. Design (lay out) input screens, output screens, and reports. ~ 8 hours
2. Design and build database (attributes, keys, indexes). ~ 4 hours
3. Design overall architecture. ~ 4 hours
4. Design program details. ~ 6 hours

III. Build the components and integrate everything into the solution.

1. Code and unit test GUI layer programs. ~ 14 hours
2. Code and unit test Logic layer programs. ~ 8 hours

IV. Perform all system-level tests and then deploy the solution.

1. Perform system functionality tests. ~ 5 hours
2. Perform user acceptance test. ~ 8 hours



WSD

WBS

To identify all tasks that must be done to complete a particular iteration and produce the defined deliverable.

WSD

To define the order of the work, e.g., the tasks in the WBS. It also helps to monitor the progress of the iteration, assign resources to the tasks to avoid duplication and plan for tasks that require extensive time or resources.

3. Discover and understand the details of the problem or the need.

Do preliminary fact-finding tasks to understand the requirements.

- → Identify key users (e.g., purchasing managers, purchasing agents)
- → Interview key users, observe existing work, review existing documentation and systems, research other companies and systems.

3. Discover and understand the details of the problem or the need.

Develop a preliminary list of use-cases and use case diagrams.

Use case: single user-triggered business event and the system's response to that event.

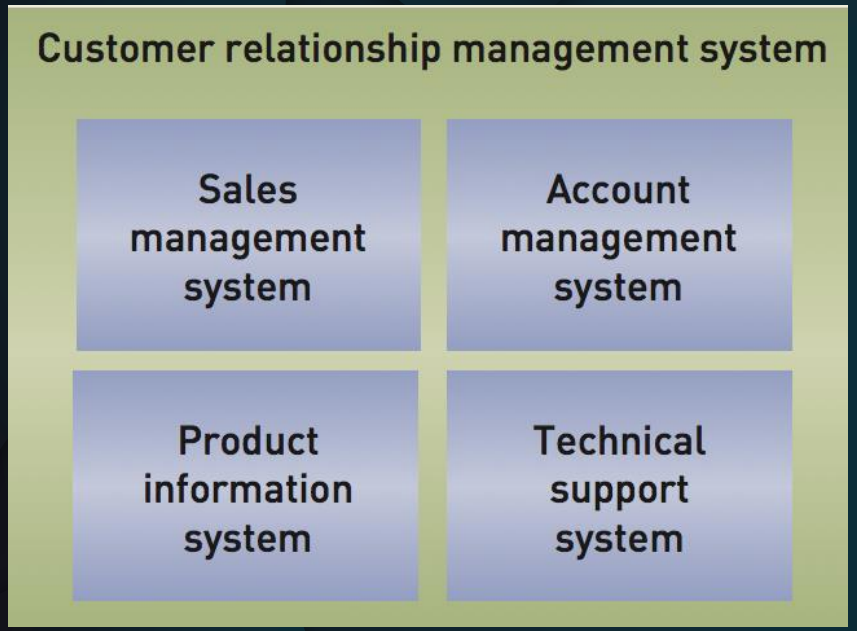
Develop a preliminary list of domain classes.

tangible items (merchandise products that you can see and touch) to more abstract concepts that you cannot touch (order). Nouns that describe categories or things.

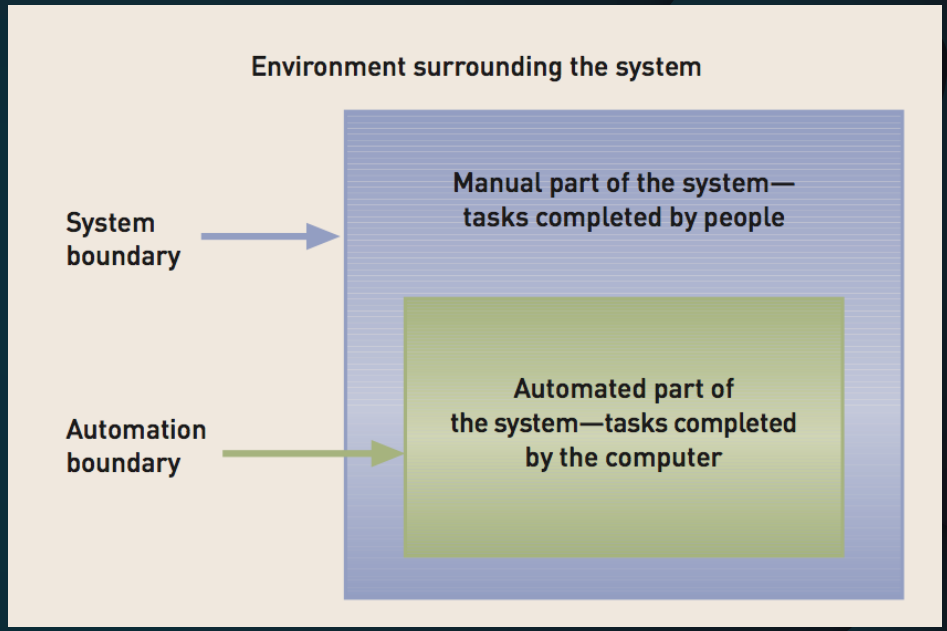
3. Discover and understand the details of the problem or the need.

Domain modelling is concerned with understanding and modelling context information for a specific problem, independently of a decision to use a software system to deal with that problem. A domain model is a representation of the main concepts in the real-world problem context.

Use Case	Description
Look up supplier	Using supplier name, find supplier information and contacts
Enter/update supplier information	Enter (new) or update (existing) supplier information
Look up contact	Using contact name, find contact information
Enter/update contact information	Enter (new) or update (existing) contact information
Look up product information	Using description or supplier name, look up product information
Enter/update product information	Enter (new) or update (existing) product information
Upload product image	Upload images of the merchandise product



Information system and subsystems

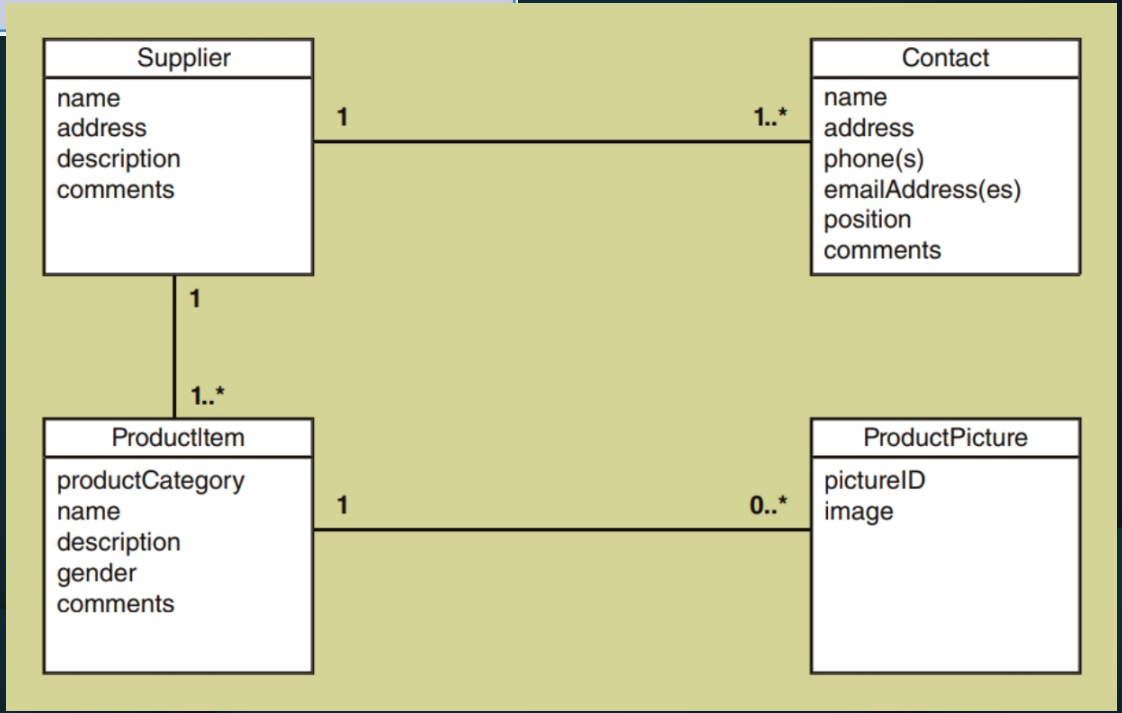


Information system boundaries

Object Classes	Attributes
Supplier	supplier name, address, description, comments
Contact	name, address, phone(s), e-mail address(es), position, comments
Product	category, name, description, gender, comments
ProductPicture	ID, image

List of object classes

Preliminary domain model class diagram



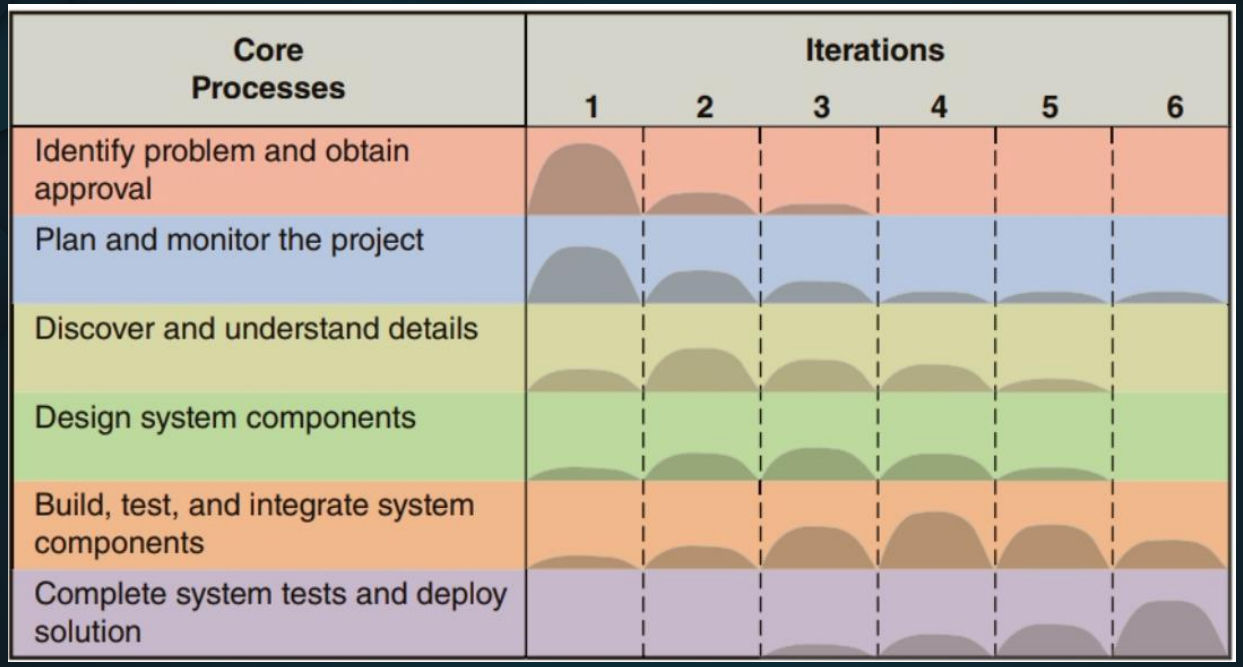
List of use cases

To identify and list the functions that must be provided by the new system.

Class diagram

To identify and list the information components that must be maintained by the new system.

Iterative development



…is an approach to system development in which the system is “grown” in an almost organic fashion.

4. Design the system components that solve the problem or satisfy the need.

Design the database structure (schema).

→ Use the domain class diagram as input and develop database schema (table design, keys, attributes)

Design the system's high-level structure: high-level design decisions will determine the detailed structure of the system.

e.g., Web-based system (more versatile) vs smartphone app (connectivity speed and control)

5. Build, test, and integrate system components.

To perform system integration testing of the developed system.

- systems integration test - the overall testing of the whole system which is composed of many sub-systems. Outcomes: all software *module dependencies* are functioning properly, and the *data integrity* is preserved between distinct modules of the whole system.
- number of additional tests: unit testing, user testing, security, accessibility, performance, stress, compatibility, etc.

6. Complete system tests and then deploy the solution.

System testing -- tests are performed to know if a complete build aligns with functional and nonfunctional requirements made for it.

System deployment -- the processes used to plan for and manage the transition of new or evolved systems and capabilities into operational use and the transition of support responsibilities to the eventual maintenance or support organization.

SYSTEM TESTING

1. To guarantee that the total build fulfills the business specifications.
2. Nonfunctional and functional type of test.
3. Black box testing.
4. Helps to identify system errors.
5. Developers and Testers

INTEGRATION TESTING

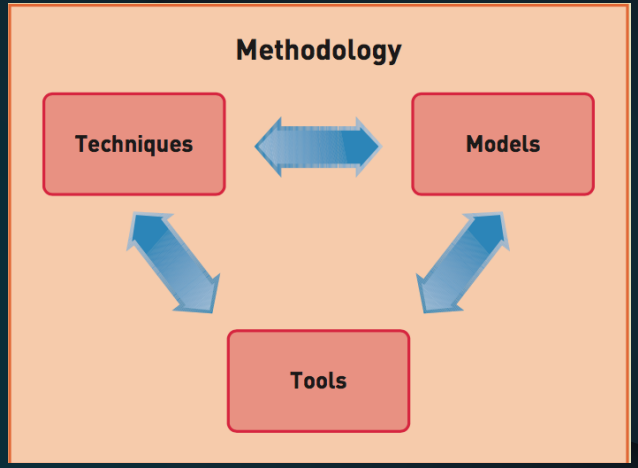
1. To guarantee that joined units can act together without problems.
2. Functional type of test.
3. White and black box testing or gray box testing .
4. Helps to identify interface errors.
5. QA

1. Intention
2. Type
3. Technique
4. Value
5. Teams involved

04

METHODOLOGY PROJECT MANAGEMENT

SYSTEM DEVELOPMENT METHODOLOGY



- models, such as diagrams, are used to describe and specify the requirements.
- techniques for how the project team will do its work.
- a set of tools—usually computer-based tools—to build models, record information, and write the code.

MODELS

- emphasizes a particular type of information
- graphical representations
- include representations of inputs, outputs, processes, data, objects, object interactions, locations, networks, and devices, among other things.

Some models of system components
- Use case diagram
- Domain model class diagram
- Design class diagram
- Sequence diagram
- Package diagram
- Screen design template
- Dialog design storyboard
- Entity-relationship diagram (ERD)
- Database schema

Some models used to manage the development process
- Gantt chart
- Organizational hierarchy chart
- Financial analysis models—NPV, payback period
- System development life-cycle model
- Stakeholders list
- Iteration plan

TOOLS

Visual modeling tools are available to systems analysts to help them create and verify important system models.

Integrated development environments (IDEs), which include many tools to help with programming tasks

- Project management application
- Drawing/graphics application
- Word processor/text editor
- Visual modeling tool
- Integrated development environment (IDE)
- Database management application
- Reverse-engineering tool
- Code generator tool

TECHNIQUES

A **technique** is a collection of guidelines that helps to complete an activity or task.

- Strategic planning techniques
- Project management techniques
- User-interviewing techniques
- Data-modeling techniques
- Relational database design techniques
- Structured programming techniques
- Software-testing techniques
- Process modeling techniques
- Domain modeling techniques
- Use case modeling techniques
- Object-oriented programming techniques
- Architectural design techniques
- User-interface design techniques



PROJECT MANAGEMENT

how to organize your work or how to manage the project.

Categorization of software projects

Successful projects, which are completed on time and within budget while meeting the users' requirements for functionality

Challenged projects, which have some combination of being late, overbudget, or reduction of scope

Failed projects, which are canceled or result in the system never being used

INTERNAL RESPONSIBILITIES

- Developing the project schedule
- Recruiting and training team members
- Assigning work to teams and team members
- Assessing project risks
- Monitoring and controlling project deliverables and milestones

EXTERNAL RESPONSIBILITIES

- Reporting the project's status and progress.
- Working directly with the client (the project's sponsor) and other stakeholders.
- Identifying resource needs and obtaining resources.

AGILE PROJECT MANAGEMENT

AGILE SCOPE MANAGEMENT

Scope isn't well understood and that there will be many changes, updates, and refinements to the requirements as the project progresses.

uncontrolled scope

AGILE TIME MANAGEMENT

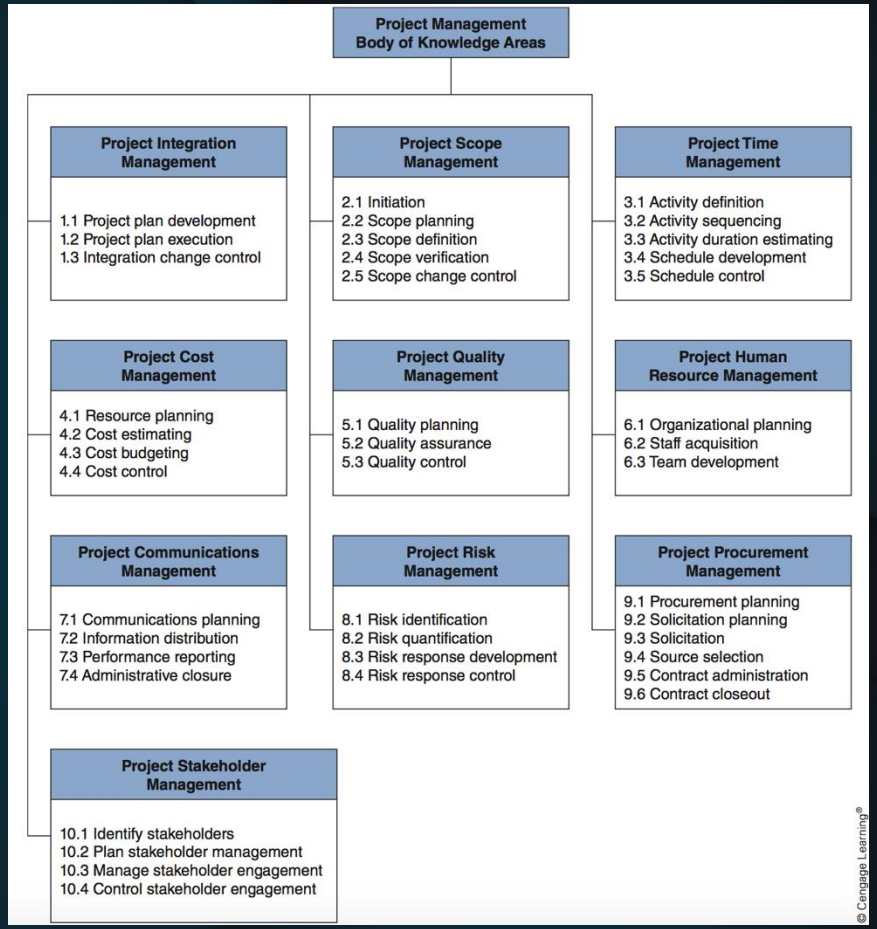
As requirements are always changing, it can be very difficult to create and maintain a meaningful project schedule.

AGILE RISK MANAGEMENT

AGILE QUALITY MANAGEMENT

- the quality of the deliverable from the project.
-the quality of the process.

PROJECT MANAGEMENT BODY OF KNOWLEDGE



Does anyone have any questions?

THANK YOU!

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